

AGODI DAVID UCHENNA

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<https://github.com/Bossmandave/>

PROFESSIONAL SUMMARY

A highly motivated and innovative Engineer with a comprehensive foundation in Mechatronics and demonstrated proficiency in developing scalable, full-stack software applications. Expertise is concentrated on the seamless integration of physical computing systems (IoT, Embedded Systems) with modern web architectures (React.js, Flask) to deliver advanced, real-world solutions for automation, security, and data management. Seeking a challenging role to leverage advanced technical competencies in software architecture, sophisticated state management paradigms, and real-time system integration.

TECHNICAL PROFICIENCIES

FRONTEND ECOSYSTEM:

React.js | Next.js | TypeScript | JavaScript | HTML5 | Tailwind CSS | Figma | Vercel | Framer motion

STATE & DATA MANAGEMENT:

Redux Toolkit | Context API | React Query | React Hook Form | RESTful API Development

BACKEND & CORE:

Python | Flask | Firebase | Authentication (Auth) | Node.js (Basic)

EMBEDDED & IOT:

Arduino | Raspberry Pi | Motor Control | Sensor Integration

TOOLS & DEPLOYMENT:

Git & GitHub | Visual Studio Code (VS Code) | Vercel | Remote Access Control | 3D Printing and Design

SELECTED PROJECTS

Facial Recognition Home Security System | *Full-Stack IoT Solution*

- Engineered a robust, access-controlled security system leveraging Raspberry Pi 4 and a USB camera to implement real-time Facial Recognition.
- Optimized data transmission and remote operational capabilities through the integration of a Flask backend, enhancing camera performance and control over the solenoid lock mechanism.
- Developed a responsive, single-page application utilizing React.js to function as a web dashboard, providing a live video feed, managing remote access permissions, and delivering instantaneous application notifications and user activity logs.

Job Listing Platform (<https://lysterpro.vercel.app/>) | *Web Development*

- Constructed a robust, full-stack digital platform facilitating job postings by employers and the management of applications by job seekers using React.js and Tailwind CSS.
- Executed the implementation of complex features, including dedicated user authentication, detailed job posting forms, and comprehensive application tracking functionalities, with state managed via advanced React Hooks.
- Ensured optimal scalability and application performance by incorporating the React Query library for highly efficient data fetching and synchronization across all user and listing data.

Certificate Verification System (<https://cverify.vercel.app/>) | *Web Development*

- Designed and deployed a secure web application infrastructure dedicated to the formal authentication and verification of documentation.
- Established a streamlined administrative dashboard for the effective management of verification records, employing a modern, clean interface built with React.js and Tailwind CSS.

Adaptive Control for Irrigation | *Mechatronics & Data*

- Conducted extensive research and developed sophisticated adaptive control algorithms to achieve dynamic optimization of water distribution, thus substantially improving water efficiency in agricultural irrigation systems.
- Conceptualized and implemented control logic designed to intelligently modulate output parameters based on the analysis of real-time environmental data inputs.

PROFESSIONAL EXPERIENCE AND ACADEMIC BACKGROUND

Professional Experience & Collaborations

- **Frontend Developer** at BrandDrive.
- **Frontend Developer Intern** at Juray Smart Solutions Ltd
- **Web Development Intern** at Zidio Web Development.
- **Robotics Intern** at Baun Robotics Educational Solutions.
- **Collaborated successfully** with 3D Robotronics on the execution of Science, Technology, Engineering, and Mathematics (STEM) projects, integrating contemporary web technologies with cloud platforms.

Education

Afe Babalola University, Ekiti State, Nigeria

B.Eng. in Mechatronics Engineering | 2020 – 2025

Relevant Coursework: Embedded Systems, Control Systems, Heat Transfer